Available online at www.sciencedirect.com
ScienceDirect
Journal of Ethnopharmacology 117 (2008) 166–169
Journal of ETHNO-PHARMACOLOGY
www.elsevier.com/locate/jethpharm

Ethnopharmacological communication

Antispasmodic activity of essential oil from *Lippia dulcis* Trev.

T. Görnemann, R. Nayal, H.H. Pertz, M.F. Melzig *

Institute of Pharmacy, Free University Berlin, Koenigin-Luise-Str. 2+4, D-14195 Berlin, Germany

Received 29 November 2007; received in revised form 2 January 2008; accepted 11 January 2008
Available online 19 January 2008

Abstract

Aim of this study: To investigate the essential oil of *Lippia dulcis* Trev. (Verbenaceae) that is traditionally used in the treatment of cough, colds, bronchitis, asthma, and colic in Middle America for antispasmodic activity.
Materials and methods: We used a porcine bronchial bioassay to study contractile responses to carbachol and histamine in the absence or presence of the essential oil.
Results: The essential oil showed anti-histaminergic and anti-cholinergic activities at 100 μg/ml.
Conclusions: The anti-histaminergic and anti-cholinergic activities of the essential oil of *Lippia dulcis* support the rational use of the plant or plant extracts to treat bronchospasm.
© 2008 Elsevier Ireland Ltd. All rights reserved.

Keywords: *Lippia dulcis*; Essential oil; Antispasmodic action

1. Introduction

1.1. Plant

Lippia dulcis Trev. (Verbenaceae) was cultivated in the greenhouse of the Institute of Pharmacy (Pharmaceutical Biology) of the Free University, Berlin. The semen of the identified plants was supplied by Prof. Mahabir P. Gupta, Facultad de Farmacia/CIFLORPAN, Universidad de Panama. Voucher specimens are present in the drug collection of the Institute of Pharmacy, Free University Berlin.

1.2. Uses in traditional medicine

Lippia dulcis Trev. (Verbenaceae) was introduced under the Nahuatl name *Tzonpelic xihuitl* ("sweet herb") by the Spanish physician Francisco Hernández in 1651 (Hernández, 1651). The plant was used by the Aztecs and Nahuatl people as a remedy for cough and urinary obstruction (Anderson and Dibble, 1961; Miranda and Valdés, 1964). In Central America, Puerto Rico, Cuba, and Mexico, this plant is employed as a decoction in the treatment of cough, cold, bronchitis, asthma, and colic (Río de la Loza, 1892; Standley, 1930; Roig y Mesa, 1945; Martínez, 1969; Díaz, 1976; Mendieta and Del Amo, 1981; Morton, 1981; Compadre et al., 1985). The Mexican Pharmacopoeia of 1884 acknowledged *Lippia dulcis* under the name of *Yerba dulce* and described the effects of the infusion as demulcent, pectoral, and emmenagogue (Maisch, 1885). The tincture of *Lippia dulcis* was used by physicians in the United States in the late 19th century in the treatment of bronchitis (Saxton, 1881; Gottheil, 1884) and acute catarrhal affections of the respiratory tract (Davis, 1890), respectively.

1.3. Previously isolated classes of constituents

The volatile oil of sweet herb has been shown to contain as main components the monoterpene camphor and the bisabolane sesquiterpenoid hernandulcin that was assessed to be more than three orders of magnitude sweeter than sucrose (Compadre et al., 1985).

* Corresponding author. Tel.: +49 30 838 51451; fax: +49 30 838 51461.
E-mail address: matthias.melzig@fu-berlin.de (M.F. Melzig).

0378-8741/$ – see front matter © 2008 Elsevier Ireland Ltd. All rights reserved.
doi:10.1016/j.jep.2008.01.009

2. Methods and materials

2.1. Plant material and extraction of oil

The aerial parts of *Lippia dulcis* were collected and dried in drying chamber at 30 °C. Aerial parts of the plants were cut and the oil was isolated by steam distillation for 4 h yielding an amount of 1.49 ± 0.11% (v/w).

GC analysis: The quantification of camphor and hernandulcin in the oil was achieved by analysing the oil by capillary GC-FID [Hewlett Packard 5890 series II (USA)], with Hp-5 (cross-linked 5% PHME siloxane) 0.5 μm coating capillary column of 30 m length and 0.32 mm ID, using the same temperature program as previously described (Souto-Bachiller et al., 1997). N_2 was used as carrier gas with a flow rate of 2.7 (ml/min). Samples (1 μl of 1% oil solutions in hexane) were injected using split-less mode by means of Auto Sampler (Hewlett Packard 7973).

2.2. Pharmacology

Lungs from adult pigs were obtained from the local slaughterhouse (Lehr- und Versuchsanstalt für Tierzucht und Tierhaltung, Teltow Ruhlsdorf, Germany). During the transportation to the laboratory, lungs were placed in ice-cold Krebs-Henseleit solution (KHS) of the following composition (mM: NaCl 118, KCl 4.7, $CaCl_2$ 2.5, $MgSO_4$ 1.2, KH_2PO_4 1.2, $NaHCO_3$ 25, and D-glucose 10, pH 7.4). The solution was aerated with 95% O_2/5% CO_2. The stem bronchus from a lower lobe was dissected free and the side branches were excised and placed in KHS. After removal of fat and connective tissue bronchial segments were cut into rings (3–5 mm long and 3–5 mm wide). Rings were horizontally suspended between two L-shaped stainless steel hooks (300 μm diameter) and mounted in water-jacketed 20-ml organ chambers filled with KHS for measurement of isometric changes in tension. The solution was kept at 37 °C and continuously aerated with 95% O_2/5% CO_2 (pH 7.4). Resting tension was adjusted to 20 mN at the beginning of the experiment. The tissues were stabilised for 60 min with replacement of the bathing medium after 30 min. During the following equilibration period (105 min) the rings were stimulated twice either with carbachol (Sigma–Aldrich, Germany; 3 μM) or histamine (Merck, Germany; 100 μM) for 20 min with a washing period of 20 min between each carbachol or histamine challenge. Resting tension was readjusted to 20 mN after the first contraction with carbachol (3 μM) or histamine (100 μM), respectively. Cumulative concentration–response curves to carbachol or histamine were constructed in the absence or presence of atropine ((Merck, Germany; muscarinic M_3 receptor antagonist), mepyramine (Sigma–Aldrich, Germany; selective histamine H_1 receptor antagonist) or the essential oil of *Lippia dulcis* (50–100 μg/ml) until a maximum response was observed. Antagonists or essential oil were added to the bathing medium 60 min before the construction of the respective concentration–response curve. Contractile effects were expressed as a percentage of the second carbachol- or histamine-induced contraction. All experiments were performed in the continuous presence of indomethacin (Sigma–Aldrich, Germany; 5 μM) to inhibit vascular eicosanoid production by cyclooxygenase. Atropine (1 μM) was additionally added in experiments with histamine to block muscarinic M_3 receptors. Control experiments were performed to verify whether vehicle (DMSO) had an influence on carbachol- or histamine-induced contractions. DMSO neither affected carbachol nor histamine-induced contractions.

2.3. Data analysis and presentation

Analytical data are presented as mean values ± S.D. for *n* experiments. Pharmacological data are presented as mean values ± standard error of the mean (S.E.M.) for separate experiments using tissues from *n* animals. Concentration–response curves were fitted to the Hill equation by an iterative least square method (GraphPad Prism 4.0, GraphPad Soft-ware, San Diego, CA, USA) to provide estimates of the maximum response E_{max} (percentage of the maximum response to a reference compound) and the half-maximum effective concentration pEC_{50} (the negative logarithm of the agonist concentration producing 50% of the maximum response). Antagonist affinities of atropine and mepyramine were expressed as an apparent pA_2 value. pA_2 was calculated from a single concentration of antagonist using the following equation: $pA_2 = -\log[B] + \log(r-1)$, where $[B]$ is the molar concentration of the antagonist and r the ratio of agonist EC_{50} determined in the presence and absence of the antagonist. Student's *t*-test (unpaired, two-tailed) was used to assess differences between two mean values with $P < 0.05$ being considered as significant.

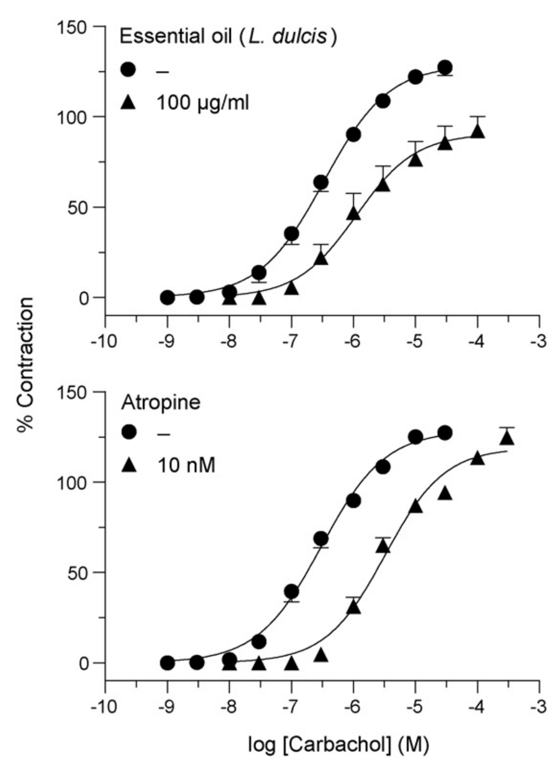

Fig. 1. Contractile responses to carbachol in porcine bronchus in the absence and presence of the essential oil of *Lippia dulcis* (upper panel) and atropine (lower panel). Contractions are expressed as a percentage of carbachol (3 μM). Points are mean values ± S.E.M. for $n = 4$ animals.

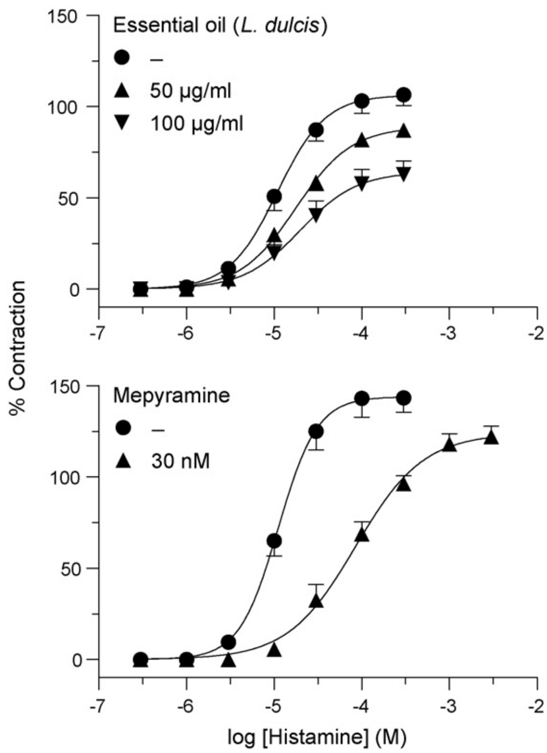

Fig. 2. Contractile responses to histamine in porcine bronchus in the absence and presence of the essential oil of *Lippia dulcis* (upper panel) and mepyramine (lower panel). Contractions are expressed as a percentage of histamine (100 μM). Points are mean values ± S.E.M. for $n=4$ animals.

3. Results

The essential oil contains as main constituents 10.1 ± 0.88% (w/v) of hernandulcin and 32.62 ± 0.18% (w/v) of camphor. The composition did not change in the experiments. The oil tastes extremely sweet with the strong after-taste of camphor.

The aim of the present study was to evaluate the antispasmodic properties of the volatile oil of *Lippia dulcis* using bronchial segments of pigs which are endowed with contractile muscarinic M_3 and histamine H_1 receptors. To establish an in vitro bioassay for these receptors located in bronchial smooth muscle, we studied the effects of carbachol and histamine in the absence and presence of the muscarinic receptor antagonist atropine and the H_1 receptor antagonist mepyramine. Porcine bronchial rings were concentration-dependently contracted by carbachol and histamine, respectively. Atropine (10 nM) and mepyramine (30 nM) antagonised the contractions produced by the respective agonist and caused a rightward shift of the concentration–response curves with no impairment of the maximal response to the agonist (Figs. 1 and 2). The calculated pA_2 values for atropine and the selective histamine H_1 receptor antagonist mepyramine were 9.01 ± 0.13 ($n=4$) and 8.28 ± 0.13 ($n=4$), respectively. In contrast to the surmountable antagonism of atropine against carbachol the essential oil of *Lippia dulcis* (100 μg/ml) failed to induce a rightward shift of the agonist curve but depressed the maximal carbachol response from 129 ± 4 to 94 ± 7% (Fig. 1). 50 μg/ml of the essential oil failed to show any effect on carbachol-induced contraction (not shown). 50 μg/ml of the essential oil failed to shift the histamine curve to the right and evoked a slight depression of the maximal histamine response from 119 ± 8 to 88 ± 3% ($P<0.05$; Fig. 2). However, 100 μg/ml of the essential oil induced a significant rightward shift of the histamine curve ($P<0.05$) and a depression of the agonist curve from 106 ± 6 to 64 ± 7%.

4. Discussion and conclusion

In the present study we demonstrated the anti-spasmodic effect of the essential oil of *Lippia dulcis*. Using atropine to block carbachol-induced contractions and mepyramine to inhibit histamine-induced contractions we could show that muscarinic M_3 receptors and histamine H_1 receptors are involved in the contractile response in this tissue. The affinity for atropine against carbachol in the present study was in line with that for the antagonist against carbachol in rabbit bronchus (Fleisch and Calkins, 1976) and against acetylcholine in porcine myometrium (Kitazawa et al., 1999). The affinity for mepyramine was in good agreement with that for the antagonist against histamine in porcine trachea (Driver and Mustafa, 1987). The obtained pharmacological results support the use of *Lippia dulcis* as traditional remedy for diseases connected with bronchospasm like cough and bronchitis. As most important component of this plant the sweet essential oil was investigated in the past (Compadre et al., 1986) but a pharmacological evaluation regarding to its use in the folk medicine was missing. The composition of the essential oil is characterized by the main components hernandulcin and camphor. It was reported that camphor has antispasmodic effects (Astudillo et al., 2004) but these results needed a confirmation by using a defined pharmacological model system. Pharmacological data on hernandulcin and the essential oil as pharmacological agent were not available in the literature. Therefore we investigated the supposed antispasmodic activity in a porcine bronchial bioassay. Our results show that the essential oil of *Lippia dulcis* produced a depression of the concentration–response curve to both carbachol and histamine with no or only a slight shift to the right. Thus the essential oil of *Lippia dulcis* behaves just like a noncompetitive antagonist of the contractile response to carbachol and histamine, respectively. It is difficult to rationalise whether the blocking properties of the essential oil of *Lippia dulcis* are due to allosteric or orthosteric inhibition (Kenakin, 2006). In any case, the essential oil of *Lippia dulcis* is able to diminish histamine H_1 and muscarinic M_3 receptor-mediated bronchoconstriction. Interestingly, the histamine action depressing effect is not only connected with a reduction of acute bronchospasm but also with an inhibition of cough (Kreutner et al., 2000). In general, H_1 receptors, which predominate in the airways of most species, mediate bronchoconstriction. H_1 receptor antagonists relieve bronchial tone and prevent histamine-induced bronchospasm (Eiser, 1983). The presence of camphor acting as expectorant (Inoue and Takeuchi, 1969) supports the antispasmodic activity. Due to the lipophilic properties of the essential oil its bioavailability might be given. The combination of both anti-cholinergic and anti-histaminergic activities in the essential oil of *Lippia dulcis* might be the reason for the traditional use of the plant for many centuries in Middle America and in former years in Europe. Because of its sweet taste extracts containing the essential oil should induce a good compliance in children and animals which are rather repelled by bitter taste.

Acknowledgement

We are grateful to Prof. Mahabir P. Gupta, Facultad de Farmacia/CIFLORPAN, Universidad de Panama for supplying the plant material.

References

Anderson, A.J.O., Dibble, C.E., 1961. Florentine Codex: General History of the Things of New Spain, Monograph 14, part 11. The School of American Research and the University of Utah, Santa Fe, New Mexico, p. 178.

Astudillo, A., Hong, E., Bye, R., Navarette, A., 2004. Antispasmodic activity of extracts and compounds of *Acalypha phleoides* Cav. Phytotherapy Research 18, 102–106.

Compadre, C.M., Pezzuto, J.M., Kinghorn, A.D., Kamath, S.K., 1985. Hernandulcin: an intensely sweet compound discovered by review of ancient literature. Science 227, 417–419.

Compadre, C.M., Robbins, E.F., Kinghorn, A.D., 1986. The intensely sweet herb, *Lippia dulcis* Trev.: historical uses, field inquiries, and constituents. Journal of Ethnopharmacology 15, 89–106.

Davis, G.S., 1890. Organic Materia Medica, 2nd edn. Parke, Davis and Co., Detroit, MI, pp. 115 and 262.

Díaz, J.L., 1976. Usos de las Plantas Medicinales de México, Monografias Cientificas II. IMEPLAN, Mexico, D.F, p. 62.

Driver, A.G., Mustafa, S.J., 1987. Correlation of histamine H_1 receptor function and [^{3}H]mepyramine binding in porcine tracheal tissue. European Journal of Pharmacology 139, 287–295.

Eiser, N.M., 1983. Histamine receptors in the bronchi. European Journal of Respiratory Diseases Supplement 128, 21–25.

Fleisch, J.H., Calkins, P.J., 1976. Comparison of drug-induced responses of rabbit trachea and bronchus. Journal of Applied Physiology 41, 62–66.

Gottheil, W.S., 1884. From Charity Hospital, New York City. Lippia Mexicana. Therapeutic Gazette 8, 58–60.

Hernández, F., 1651. Rerum Medicarum Novae Hispaniae Thesaurus seu Plantarum, Animalium, Mineralium Mexicanorum Historia. Typographeio Vitalis Mascardi, Rome, p. 240.

Inoue, Y., Takeuchi, S., 1969. Expectorant-like action of camphor derivatives. Nippon Ika Daigaku Zasshi 36, 351–354.

Kenakin, T.P., 2006. A Pharmacology Primer, 2nd edn. Academic Press, Amsterdam.

Kitazawa, T., Uchiyama, F., Hirose, K., Taneike, T., 1999. Characterization of the muscarinic receptor subtype that mediates the contractile response of acetylcholine in the swine myometrium. European Journal of Pharmacology 367, 325–334.

Kreutner, W., Hey, J.A., Anthes, J., Barnett, A., Young, S., Tozzi, A., 2000. Preclinical pharmacology of desloratadine, a selective and nonsedating histamine H_1 receptor antagonist. Arzneimittelforschung 50, 345–352.

Maisch, J.M., 1885. On some useful plants of the natural order of Verbenaceae. American Journal of Pharmacy 57, 330–335.

Martínez, M., 1969. Las Plantas Medicinales de México, 5th edn. Ediciones Botas, Mexico, D.F, pp. 166–167.

Mendieta, R.M., Del Amo, S., 1981. Plantas Medicinales del Estado de Yucatán. Compañia Editorial Continental, Mexico, D.F, p. 199.

Miranda, F., Valdés, J., 1964. Comentarios Botánicos. In: del Pozo, E.C. (Ed.), Martin de la Cruz. Libellus Medicinalibus Indorum Herbis. Instituto Mexicano del Seguro Social, Mexico, D.F., pp. 243–284.

Morton, J.F., 1981. Atlas of Medicinal Plants of Middle America. Bahamas to Yucatan. Charles C. Thomas, Springfield, IL, pp. 746–747.

Río de la Loza, M., 1892. Yerba dulce. La Farmacia (Mexico) 1, 3–7.

Roig y Mesa, J.T., 1945. Plantas Medicinales, Aromáticas o Venenosas de Cuba, Part 1. Ministerio de Agricultura, Servicio de Publicidad y Divulgación, Havana, pp. 508–509.

Saxton, J.H., 1881. Lippia Mexicana. Therapeutic Gazette 7, 3.

Souto-Bachiller, F.A., De Jesus-Echevarría, M., Cárdenas-González, O.E., Acuña-Rodriguez, M.F., Meléndez, P.A., Romero-Ramsey, L., 1997. Terpenoid composition of *Lippia dulcis*. Phytochemistry 44, 1077–1086.

Standley, P.C., 1930. Flora de Yucatán, Botanical Series, vol. III, no. 3, Publication 279. Field Museum of Natural History, Chicago, IL, p. 402.